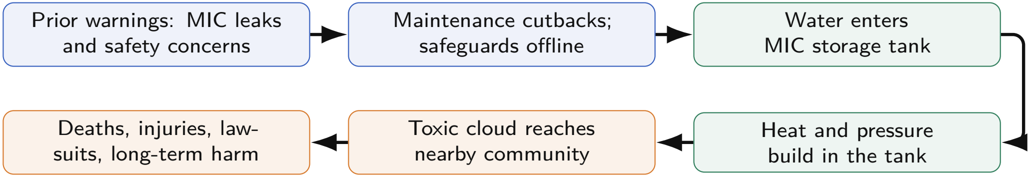
Lecture 3: Understanding Ethical Problems

HUM 4441: Engineering Ethics

Reaz Hassan Joarder, Samnun Azfar, and Yousuf Ibne Kamal Niloy
Junior Lecturer
{reazhassan, samnunazfar, yousufniloy}@iut-dhaka.edu
Department of CSE
Islamic University of Technology

Bhopal: The Night the Safeguards Failed



In December 1984, methyl isocyanate gas escaped from a Union Carbide plant in Bhopal, India. The ethical problem is the connection between dangerous technology, organizational choices, and public vulnerability. *In this case, "technical failure became moral failure".*

(Fleddermann, Ch. 3, pp. 37, 50–51)

The ethical question

How should engineers reason when economic benefit, technical uncertainty, public danger, and organizational responsibility collide?

The Problem Is Larger Than the Leak

- *Hazardous storage* created high-consequence risk.
- *Cost cutbacks* disabled key safeguards.
- *Refrigeration, alarm, flare, and scrubber layers* failed.
- *Dense nearby settlement* amplified exposure.
- *Weak warning and evacuation planning* worsened harm.

First discussion: Which part of the Bhopal case looks most like an engineering responsibility rather than a random accident?

Why Ethical Theories Belong in Engineering

Engineering theory

- *Defines concepts* used in the problem.
- *Organizes evidence* and assumptions.
- Gives a *disciplined method* for judgment.

Moral theory

- *Defines what counts morally*.
- Makes *harms, duties, rights, and character* visible.
- Lets disagreement be *analyzed*, not shrugged off.

Same role, different domain

Both kinds of theory turn a messy situation into an analyzable problem. In ethics, the *"loads"* are harms, duties, rights, consequences, and character.

A Short History of Ethical Thought

Ancient religious moral law → Greek ethical philosophy → Modern secular moral theory → Law, rights, and professional codes

Western ethical thought draws from religious traditions, Greek philosophy, later thinkers such as Locke, Kant, and Mill, and legal codification.

Engineering takeaway

Ethical conduct is grounded in concern for other people, not only in law, religion, custom, or company policy.

What a Moral Theory Does

A moral theory defines terms in a uniform way and links ideas together so problems can be analyzed consistently.

- It helps identify *morally relevant facts*.
- It makes *hidden assumptions* visible.
- It gives a shared language for *ethical disagreement*.

Lens	Main concern
Utilitarianism	*Overall welfare*
Duty ethics	*Obligations*
Rights ethics	*Moral claims of persons*
Virtue ethics	*Character and habits*

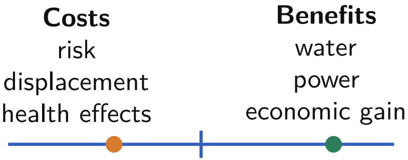
Four Lenses for One Decision

Lens	Core question
Utilitarianism	What produces the *best overall consequences* for everyone affected?
Duty ethics	What *obligations* must be honored even when payoff is inconvenient?
Rights ethics	Whose *rights* are at stake, and what would violate them?
Virtue ethics	What kind of *professional character* does this action express?

Use the four lenses as a *checklist for analysis*, not as four equally final answers or a *"vote"*.

Utilitarianism: Consequences for Everyone Affected

- The goal is *not private gain*.
- The unit of concern is *society as a whole*.
- Engineering versions include *risk–benefit* and *cost–benefit* reasoning.



Alert

Main danger: total benefit can hide a concentrated burden. Ask whose welfare is being counted, and whose welfare is being discounted.

Utilitarianism in Engineering: Dam and WIPP

A dam project

May provide *drinking water, flood control, and recreation* while displacing people and changing ecosystems.

WIPP

May solve a national nuclear-waste problem while putting *transport-route communities* under accident risk.

Ask	Why it matters
Benefits	Often *broad and visible*
Costs	Often *local and concentrated*
Voice	Usually *unequal among affected groups*
Unknowns	Consequences still require *prediction under uncertainty*

Discussion: Who benefits, who pays, and who gets to decide?

Cost–Benefit Analysis: Useful, But Not Enough

What CBA does	Effect
Clarifies	*Cost, feasibility, expected gains, and trade-offs*
Can hide	*Dignity, fairness, ecology, fear, trust, unequal burden*
Cannot decide	Whether available alternatives are *ethical in the first place*

Cost–benefit analysis is an engineering planning tool. It can help compare ethical alternatives, but it should not be used as a substitute for ethical analysis. *It informs judgment; it does not replace judgment.*

Decision rule

First identify ethically acceptable options; then compare cost and feasibility among those options.

Duty and Rights Ethics: Respect for Persons

Duties

- *Be honest*.
- *Avoid preventable harm*.
- *Treat others fairly*.
- Respect *autonomy and consent*.

Rights

- *Life and safety*
- *Liberty*
- *Property*
- *Privacy and informed choice*

Core protection

A good outcome for many people does not automatically justify exposing particular people to serious harm without respect, consent, or protection.

When Rights Collide

A public project may protect

- *Access to water*
- *Flood control*
- *Reliable infrastructure*
- *Public health*

The same project may threaten

- *Homes and property*
- *Local ecosystems*
- *Community identity*
- *Meaningful consent*

Rights language clarifies what is **morally at stake**, but it does not automatically decide which claim has priority.

Virtue Ethics: The Character of Practice

Virtues in engineering

- *Honesty*
- *Competence*
- *Responsibility*
- *Fairness*
- *Trustworthiness*

Questions to ask

- Is this action *honest*?
- Does it show *competence* or convenient ignorance?
- Would it still look honorable if made *public*?

Professional character

Personal morality and professional morality should not be split into separate lives.

Be Careful With Virtue Words

Helpful virtues

Honesty, responsibility, competence, fairness, trustworthiness, caring, citizenship, respect.

Vices

Dishonesty, irresponsibility, incompetence, disloyalty to the public, avoidable ignorance.

Words to inspect

Honor, loyalty, pride, reputation, and team spirit can become dangerous when they excuse silence or cover-ups.

Virtue ethics is useful only when the named virtue actually supports good professional conduct, not just *"team reputation"*.

Can a Corporation Be Moral?

Three levels

- *People* make decisions.
- *Organizations* create systems.
- *Communities* experience consequences.

A corporation is not a person in the ordinary moral sense, but corporate action still affects persons. Responsibility should not disappear behind an organization chart or behind *"the system made us do it"*.

Accountability

For ethical analysis, corporations should be treated as accountable actors while individuals still carry role-based responsibility.

Which Theory Should We Use?

A practical method

1. *State the ethical problem*.
2. Analyze it with *all four theories*.
3. Compare *agreements and tensions*.
4. Give serious weight to *rights and duties*.
5. Form a *balanced judgment*.

- Different theories often reach the *same conclusion*.
- When they conflict, do not *average them mechanically*.
- Defend judgment using the *evidence each lens reveals*.

Bhopal Through Four Lenses

Lens	**Reading of the case**
Utilitarianism	Economic benefit cannot justify *catastrophic chemical risk* without reliable safeguards.
Duty and rights	The public had a claim against preventable danger; the company had duties of maintenance, warning, and emergency planning.
Virtue ethics	*Diligence, competence, and honesty* are violated when leaks and broken safety systems become normal.
Corporate morality	Responsibility spreads across design, management, training, regulation, and knowledge transfer.

Ethics Is Not Geography

The foundations differ across traditions, but engineering practice often converges around **honesty**, **competence**, **safety**, **fairness**, **respect**, and care for the community.

Cultural humility matters, but ethics cannot be reduced to local taste or *"whatever this workplace usually does"*.

Chinese and Indian Ethical Traditions

Chinese traditions

- *Practical moral cultivation*
- *Ideal character traits*
- *Mutual respect*
- *Interdependence* between individual and community

Engineering connection: balance public benefit with respect for affected individuals.

Indian traditions

- *Practical pursuit* of a good life
- *Truth and compassion*
- *Self-control and gentleness*
- *Freedom from greed*

Engineering connection: professional service requires restraint and attention to human consequences.

Muslim and Buddhist Ethical Traditions

Muslim ethics

- *Virtue-oriented* moral life grounded in the Qur'an
- *Humility, honesty, kindness, trustworthiness*
- Professional relevance: *truthful claims* and fair dealing

Buddhist ethics

- Avoid *destruction of life, theft, lying, and harmful excess*
- Cultivate *learning, diligence, generosity, patience, and limits*
- Professional relevance: *safety, sustainability, and competence*

Why Codes Converge Across Cultures

Different foundations

Religious, philosophical, and cultural origins vary across societies.

Shared risks

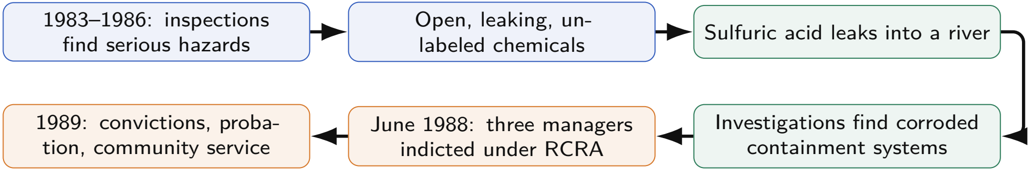
Bridges, software, chemicals, power systems, and data systems can harm people anywhere.

Similar duties

Codes converge around safety, honesty, competence, fairness, and public welfare.

Reflection: What ethical value would you expect to find in any engineering code, regardless of country?

The Aberdeen Three



Three civilian engineer-managers were convicted after hazardous-waste storage and disposal failures at the Pilot Plant. Responsibility followed from authority over storage practices and safety equipment: *"authority creates accountability"*. (Fleddermann, Ch. 3, pp. 51–52)

Case lesson

Engineering responsibility includes maintenance, supervision, and control of dangerous conditions.

Aberdeen Through the Ethical Lenses

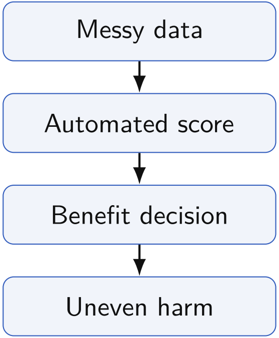
Lens	**Reading of the case**
Utilitarianism	Unsafe storage created risks to workers, the environment, and nearby water systems for little defensible benefit.
Duty and rights	Managers had duties to know and control hazardous materials; others had rights not to be exposed to *preventable danger*.
Virtue ethics	Competence and responsibility require more than saying *"this is how we have always done it"*.

Case question: Does it matter ethically whether the unsafe choices were engineering decisions, management decisions, or both?

A Computing Version of the Same Problem

A city deploys an automated system to prioritize public benefits applications. It promises faster decisions and lower cost. Early testing shows higher error rates for applicants whose records are incomplete or inconsistent.

- *The benefit is real*.
- *The harm is unevenly distributed*.
- The public may not understand the system well enough to *contest it*.



Discussion: Analyze this first as a utilitarian, then as a rights ethicist. What changes?

Teamwork Has an Ethical Dimension

Failures that hurt the team

- *Missing assigned work*
- *Hiding delays or errors*
- *Withholding information*
- *Taking over so others cannot contribute*

Virtues that support the team

- *Reliability*
- *Openness*
- *Fairness*
- *Shared responsibility*
- *Support for others' learning*

Reflection: In a software project, when does "I will just do it myself" become an ethical problem rather than efficiency?

Fleddermann, Ch. 3, p. 52

Lecture 3 Summary

- Ethical theories give engineers *disciplined lenses* for harm, benefit, duty, rights, character, and responsibility.
- Utilitarianism and cost–benefit reasoning are powerful, but they can hide *uneven burdens* and hard-to-price losses.
- Duty and rights ethics protect persons from being treated only as "*means*" to an aggregate goal.
- Virtue ethics asks what kind of *professional character* a decision expresses and reinforces.
- Bhopal and Aberdeen show why ethical analysis must include *systems, maintenance, warning, authority, and public impact*.
- Ethical traditions differ in origin, but practice converges around *honesty, competence, safety, fairness, and respect*.